Statistics 8330: Elements of Statistical Learning

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Office Hours:

Monday 2-3
Wednesday after class
Zoom by appointment at https://zoom.us/j/92913522978

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

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Course Goals:
This course explores advanced statistical learning techniques, covering nonlinear methods, dimension reduction, classification, clustering, ensemble methods, and model evaluation. Use of simulations for evaluating, comparing, and understanding methods. Emphasizing practical applications, students will fit models using Python and R to solve real-world data science challenges.

After successful completion of this course, students will be able to:

1. **Differentiate** between supervised and unsupervised learning approaches and **analyze** the bias-variance tradeoff in model selection.
2. **Assess** model performance using cross-validation, performance metrics, and regularization techniques to prevent overfitting and improve generalization.
3. **Apply** nonlinear modeling techniques, including polynomial regression, splines, and kernel methods, to capture complex relationships in data.
4. **Implement** dimension reduction techniques such as PCA, factor analysis, and manifold learning to improve model interpretability and efficiency.
5. **Compare and evaluate** classification and clustering methods, including logistic regression, SVM, k-NN, k-means, and Gaussian mixture models, for various data scenarios.
6. **Construct** ensemble learning models, including bagging, boosting, and stacking, to enhance predictive accuracy and model robustness.
7. Use **computer simulations** to assess and compare statistical methods.
8. **Apply** statistical learning techniques in R and Python to **analyze** real-world datasets and **communicate** findings effectively through written reports.

Topical Outline

1. **Introduction to Statistical Learning**
    - Overview of supervised vs. unsupervised learning
    - Bias-variance tradeoff and model complexity
    - Key concepts in statistical learning theory
2. **Monte Carlo Simulations**
    - Formulating the assumptions of a simulation.
    - Implementing simulations in computer programs.
    - Using simulations to assess statistical methods.
3. **Model Assessment and Evaluation**
    - Cross-validation and resampling methods
    - Performance metrics (AUC, F1-score, log-loss)
    - Overfitting, regularization, and interpretability
4. **Nonlinear Methods**
    - Polynomial regression and splines
    - Generalized additive models (GAMs)
    - Kernel methods and smoothing techniques
5. **Dimension Reduction Techniques**
    - Principal Component Analysis (PCA)
    - Lasso
    - Manifold learning (t-SNE, UMAP)
6. **Classification Methods**
    - Logistic regression and discriminant analysis
    - Support vector machines (SVM) and kernel methods
    - k-Nearest Neighbors (k-NN)
7. **Clustering Methods**
    - k-Means and hierarchical clustering
    - Model-based clustering and Gaussian Mixture Models (GMM)
    - Spectral clustering and latent variable models
8. **Ensemble Methods**
    - Bagging and boosting
    - Random forests and gradient boosting machines
    - Stacking and blending strategies
9. **Practical Applications**
    - Applications will be emphasized throughout the course.
    - Case studies in healthcare, finance, marketing, and other areas.
    - End-to-end model deployment in Python and R
    - Ethical considerations and model fairness

Prerequisite: Stat 6420. I do not recommend taking this course unless your statistical knowledge is at the level of this course or higher. I am also assuming that you have basic knowledge of R. Talk to me if you have doubts.

Textbook:

The Elements of Statistical Learning. Trevor Hastie, Robert Tibshirani, and Jerome Friedman.

The book can be downloaded at

Elements of Statistical Learning: data mining, inference, and prediction. 2nd Edition.

Another useful book is

An Introduction to Statistical Learning with Applications in R/Python. D. Witten, G. James, R. Tibshirani, and T. Hastie (Springer).

This book is a lower-level version of the Elements of Statistical Learning. The book can be downloaded at

An Introduction to Statistical Learning (statlearning.com)

Note that there are versions for both R and Python!

Grading: 80% homework, 20% in-class final. Note: these percentages are approximate; I will be the final judge of your grade.

As a University of Georgia student, you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in "A Culture of Honesty" found at: www.uga.edu/honesty. Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Note on homework: The homework assignments make up 80% of your grade. There will be 6-8 assignments. Thus, each one will worth around 10-13% of your final grade and missing even a single one will move your final grade by a full letter grade or more(e.g. A to B). DO NOT MISS ANY ASSIGNMENTS IF YOU WANT TO GET A GOOD GRADE.

Final Exam: The final exam will be in class and will cover the whole semester. It will be open book and open notes. It will be at the university scheduled final exam time for this class: **Monday May 4 from 8:00-11:00.**

Late Policy: Unless you make prior arrangements, there will be a 10% late penalty if assignments are not turned in by midnight of the day that they are due. There will be an additional 20% for each additional 24 hours that they are late (that is, the penalty is incurred at midnight each day). If there is a valid reason (illness, travel, etc) that you will not be able to turn in the assignment on time, email me about an extension. Extensions **must be approved ahead of a time** and must be requested no later than 24 hours in advance.

Homework assignments and exams will be submitted on eLC. It is your responsibility to make sure that everything is correctly submitted on eLC. This includes making sure that it was correctly uploaded and that files are complete and are readable. If it isn't submitted correctly by the deadline, then it didn't meet the deadline.

eLearning Commons: I will use eLearning Commons for distribution of course materials. The website is https://www.elc.uga.edu. All UGA students have an eLC account and you should see this course when you log in.

Guidelines for homework grading: Assignments in this class will generally require you to write an R or Python program to conduct some data analysis or simulation task. If your program does not run due to syntax errors, it will be returned to you to fix and there will be a point deduction of at least 25% (this does not apply to simple errors arising from translations from your computer to the TA's or mine). If the program is not substantially correct (i.e. the basic idea isn't there), there will be a point deduction of at least 50%. If the program is giving results that are easily seen to be incorrect, there will be a point deduction of at least 25%. The message here is that you shouldn't hand in shoddy work. I am usually willing to give extensions on assignments as long as people don't make a habit of turning assignments in late. I do reserve the right to stop giving extensions if I feel that people are abusing it.

Computers: We will be doing frequent programming work in class. You should bring your laptop to class.

Software Installation

You should install the following software on your computer:

For Python, we will use Anaconda. You can download this at
Download Anaconda Distribution | Anaconda

Just download, double click, and follow the instructions,

You can download R from the site http://cran.r-project.org/.

RStudio gives an improved user interface to R. You can download this at
https://www.rstudio.com/products/rstudio/download/
Again, just download, double click, and follow the instructions.
Let me know if you have any questions or problems.

Outside Resources and ChatGPT

Everyone who codes uses the internet to figure out how to do things. Knowing how to look things up is a necessary skill for a programmer. However, there is a difference between using the internet as an occasional aid to programming and using it as a substitute for programming. Too many students spend hours googling for a solution rather than figuring out how to do it themselves. Coding is something that you learn by doing, not by googling. Spend your time trying things for yourself. If it doesn't work, figure out why it doesn't work and then fix it. A large part of learning coding and math is learning the mental skills and the mental fortitude to keep hammering at a problem until you solve it. If you solve the problem yourself, you will understand it and you will know what to do when you encounter it in the future. If you find the solution on the internet, you probably won't understand it and you also won't be able to tell whether it is actually correct.

This issue is magnified by ChatGPT and other large language models (LLMs). ChatGPT can write code for you. Sometimes it does brilliantly. However, it also gets it completely wrong on a regular basis. The internet is full of examples of ChatGPT and other LLMs getting things completely (and often hilariously) wrong.

I have experimented with using ChatGPT with coding problems similar to what we do in this class. It produces code that is wrong a substantial percentage of the time. A lot of people in the programming world have been experimenting with ChatGPT and the consensus seems to be that it usually does well with simple things, but often makes mistakes with more complex things. Even with the simple things it makes mistakes often enough that you cannot depend on it. Furthermore, it is not easy to tell that is wrong. It produces nice looking code that runs and produces output that often seems reasonable. In some cases I was only able to tell that it was wrong because I had working code to compare it to. See this study, which found that ChatGPT gets coding questions wrong more than half of the time:

https://arxiv.org/abs/2308.02312

The bottom line is that depending on ChatGPT to write code is a bad idea: it will give you bad results much of the time.

The more important reason to not use ChatGPT for this class is because if you do you won't learn the skills that you need. We are at the beginning of what is likely to be a major shift in how people work in many fields and no one knows what the landscape will be in two years, let alone ten or twenty. There is no doubt that LLMs will become an important part of the programmer's toolkit and you should be learning to use them. However, you need to learn to think and solve problems on your own. If all that you can do is type things into ChatGPT, then what value do you add? Why would anyone hire you? If you want to be valuable in the AI world, then AI needs to be part of your toolkit but not a substitute for your own brain. The only way that you can learn to use your brain effectively in the context of programming is to learn to program. The only way to do that is to spend a lot of time figuring out how to program on your own.

In this spirit **Using LLMs and other similar tools for assignments is not permitted in this class unless I specify otherwise.** I know that many of you will be tempted to use it and that this ban will be hard to enforce. However, I will do my best to enforce it and using LLMs on an assignment will be grounds for a zero grade for that assignment. I will structure assignments in ways that make using LLMs more difficult and we will also use tools to detect AI. Finally, I will repeat again: I have tested it and ChatGPT gives incorrect code for many assignments in this class. It will fail you if you depend on it.

UGA Student Honor Code

"I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others." A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at honesty.uga.edu. Every course syllabus should include the instructor's expectations related to academic honesty.

UGA Well-Being Resources:

UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community.

Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit sco.uga.edu. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity.

UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

- *Well-being Resources: well-being.uga.edu*
- *Student Care and Outreach: sco.uga.edu*
- *University Health Center: healthcenter.uga.edu*
- *Counseling and Psychiatric Services: caps.uga.edu or CAPS 24/7 crisis support at 706-542-2273*
- *Health Promotion/ Fontaine Center: healthpromotion.uga.edu*
- *Accessibility and Testing:* accessibility.uga.edu

  Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting https://well-being.uga.edu.